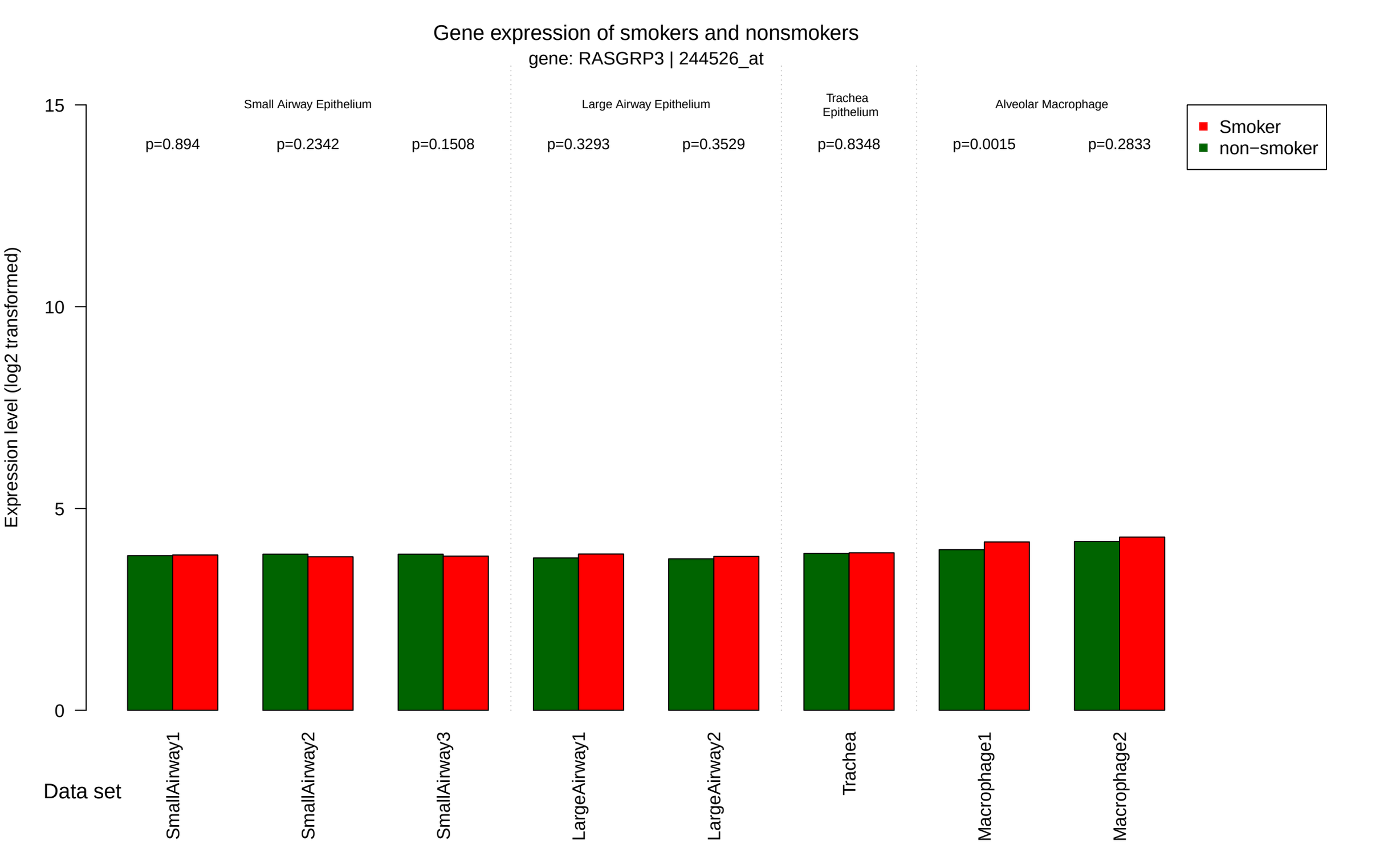
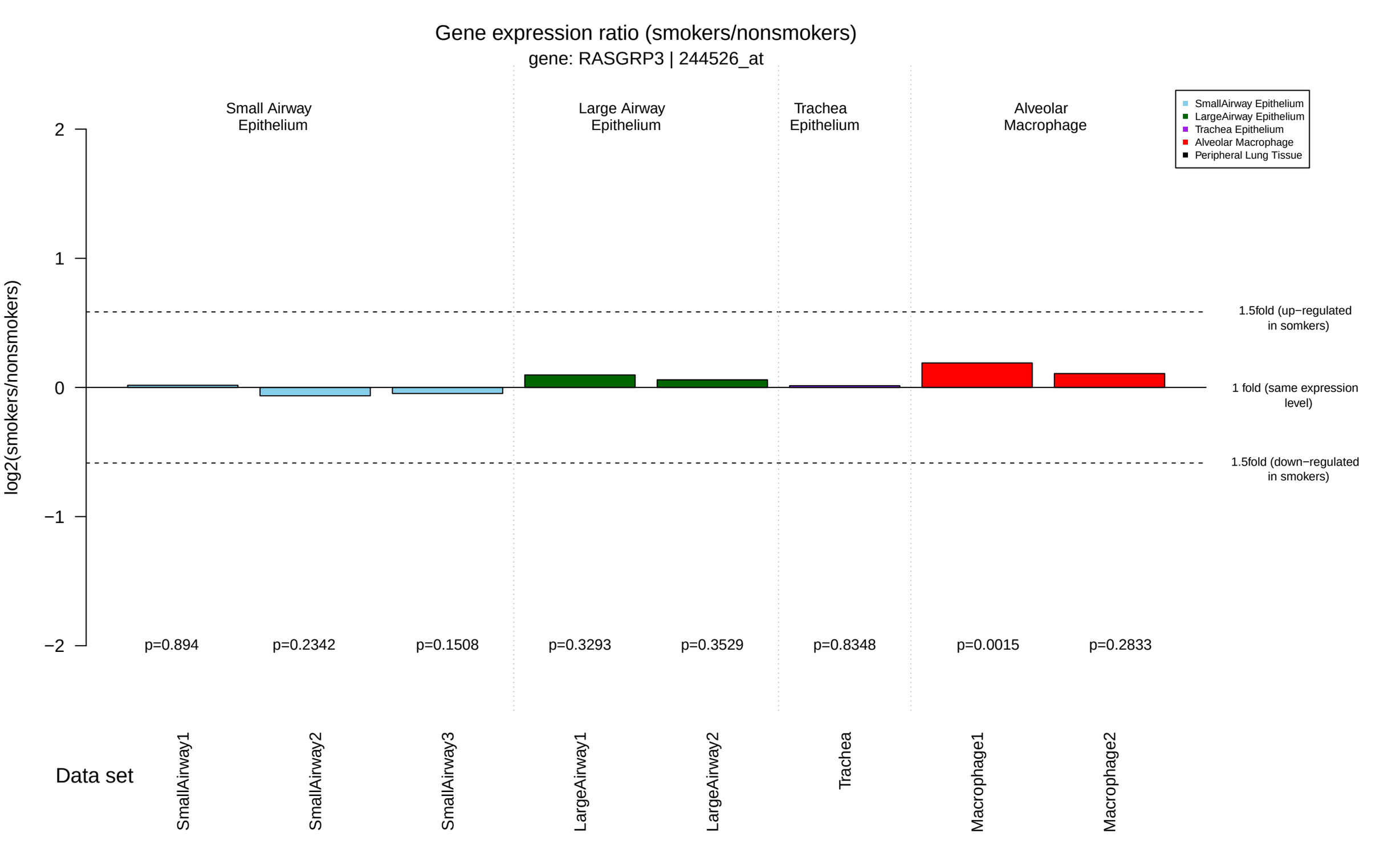
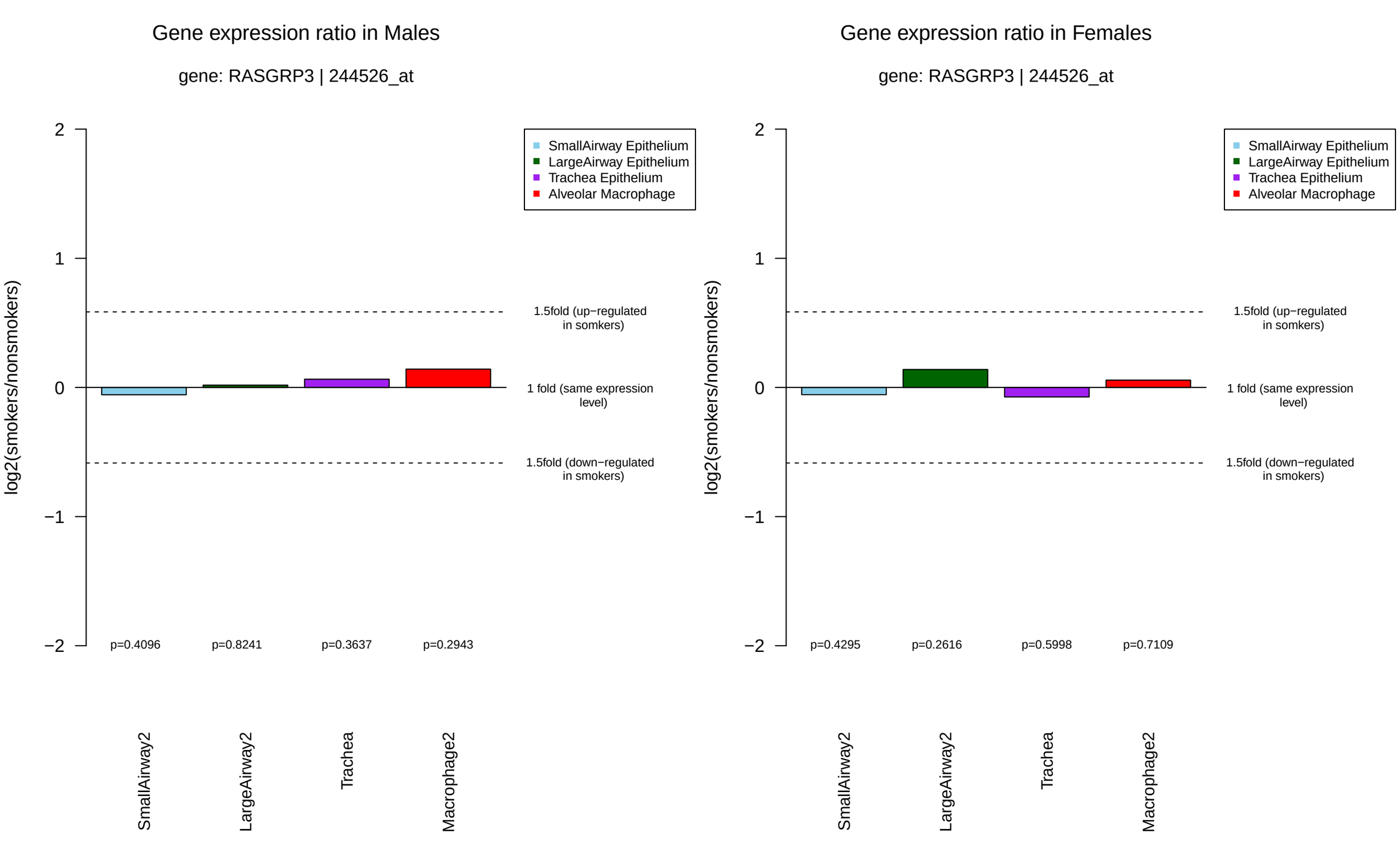
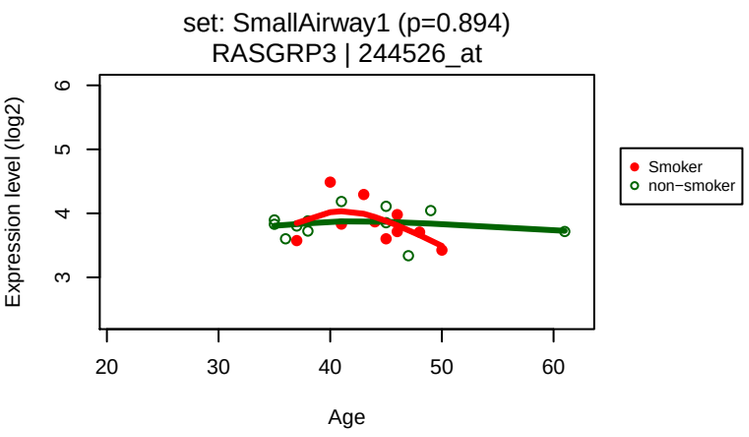
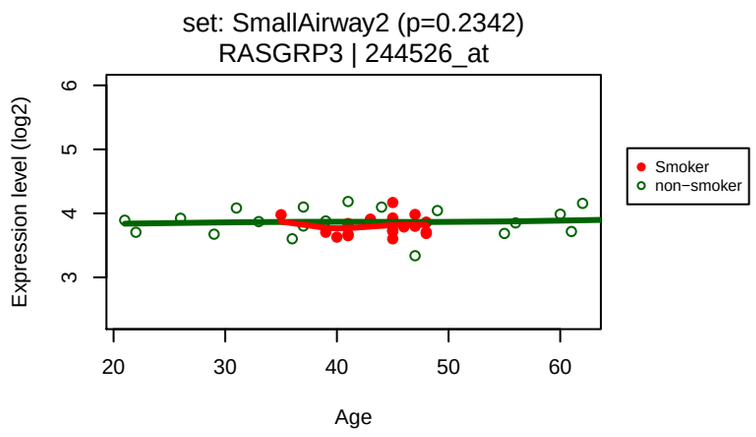
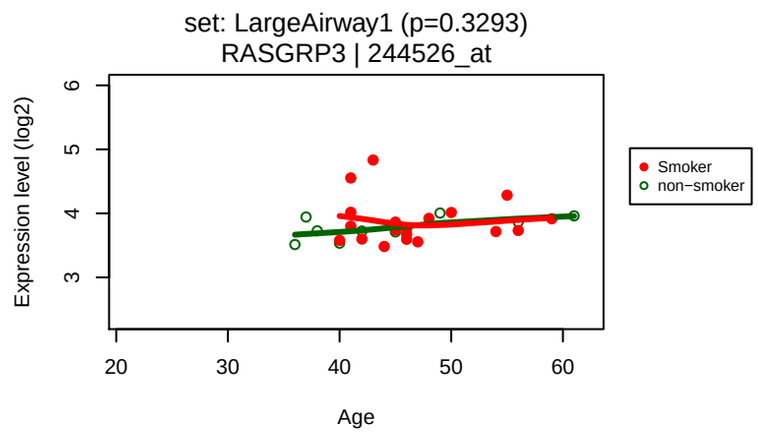

Gene expression of smokers and nonsmokers
gene: RASGRP3 | 244526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.894
p=0.2342
p=0.1508
p=0.3293
p=0.3529
p=0.8348
p=0.0015
p=0.2833
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

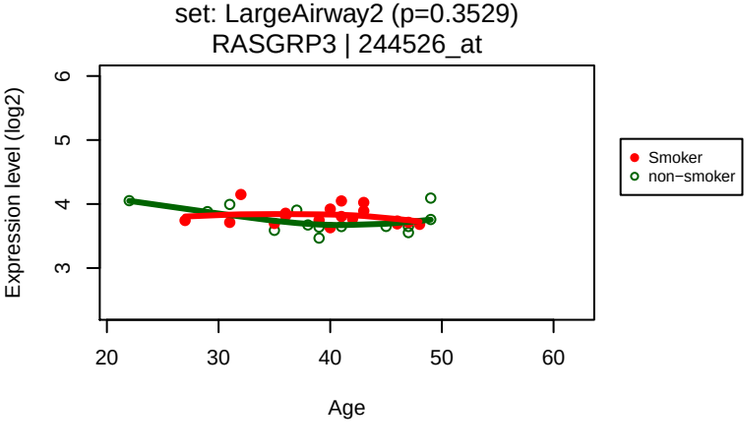

Gene expression ratio (smokers/nonsmokers)
gene: RASGRP3 | 244526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.894
p=0.2342
p=0.1508
p=0.3293
p=0.3529
p=0.8348
p=0.0015
p=0.2833
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

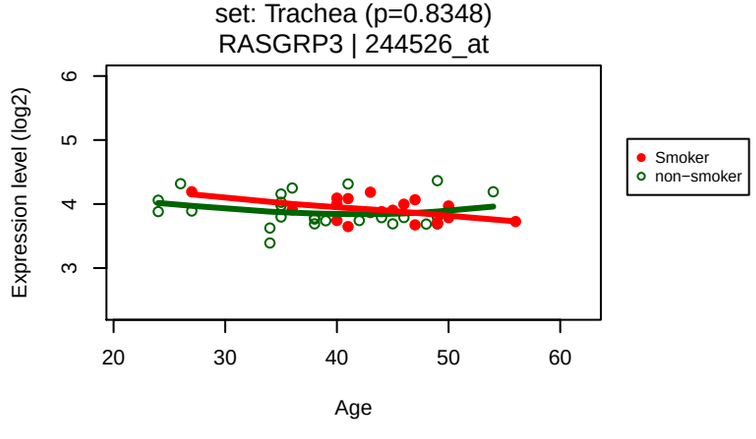

Gene expression ratio in Males
gene: RASGRP3 | 244526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4096
p=0.8241
p=0.3637
p=0.2943
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGRP3 | 244526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4295
p=0.2616
p=0.5998
p=0.7109
SmallAirway2
LargeAirway2
Trachea
Macrophage2

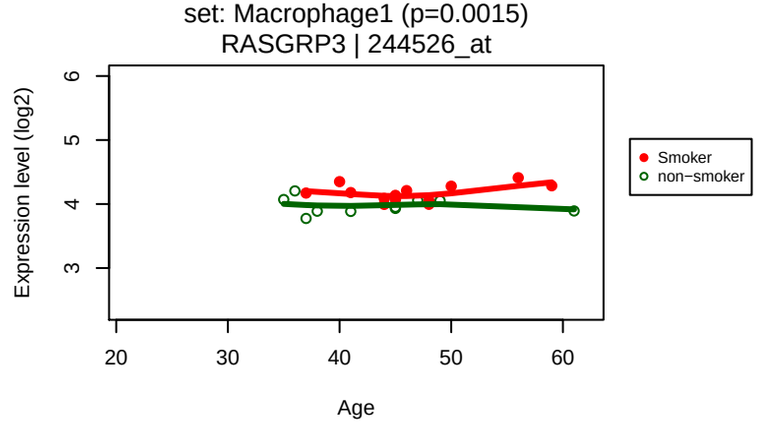

set: SmallAirway1 (p=0.894)
RASGRP3 | 244526_at
Expression level (log2)
Age
Smoker
non−smoker

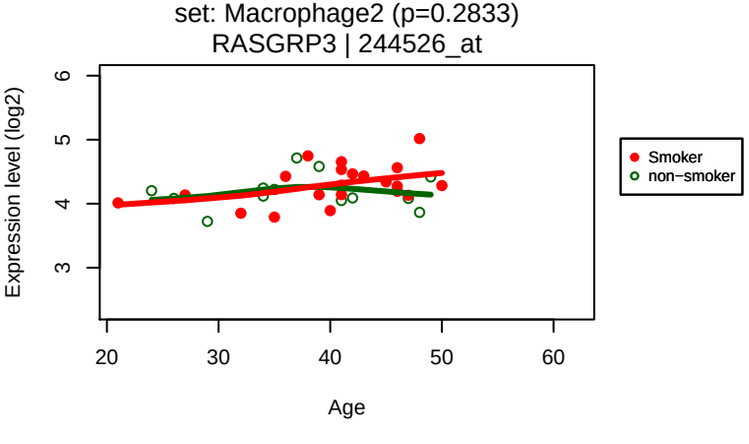

set: SmallAirway2 (p=0.2342)
RASGRP3 | 244526_at
Expression level (log2)
Age
Smoker
non−smoker

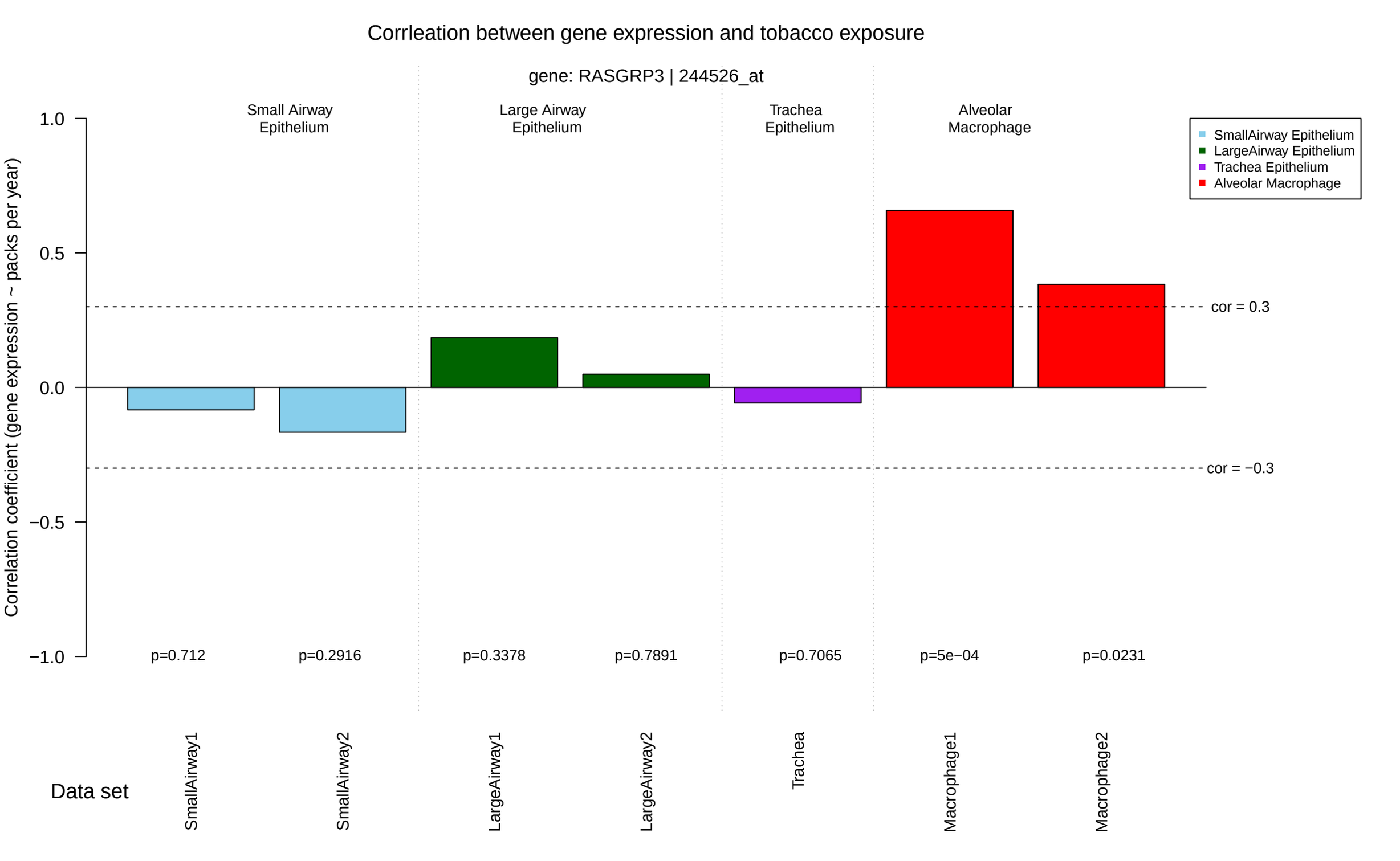

set: LargeAirway1 (p=0.3293)
RASGRP3 | 244526_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3529)
RASGRP3 | 244526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8348)
RASGRP3 | 244526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0015)
RASGRP3 | 244526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2833)
RASGRP3 | 244526_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGRP3 | 244526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.712
p=0.2916
p=0.3378
p=0.7891
p=0.7065
p=5e−04
p=0.0231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2